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2027 CYCLE

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CUET UG

Physics (322)

Electromagnetic Waves

Official Syllabus Unit 5 · Class 12 NCERT · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA Physics-322, Unit 5)	Key Exam Focus & Skills
Displacement Current	Concept of displacement current.	Inconsistency of Ampere's circuital law, Maxwell's correction $I_d = \varepsilon_0\left(d\frac{\Phi_E}{d}t\right)$, current continuity $I_c = I_d$ across capacitor boundaries, magnetic field inside capacitor gap.
Maxwell's Four Equations	Electromagnetic waves, their characteristics, and their transverse nature.	Gauss's laws for (E) and (B), Faraday's law of induction, Ampere-Maxwell law, source of electromagnetic waves (accelerated electric charges).
Transverse Propagation & Speed	Electromagnetic waves, their characteristics, and their transverse nature.	Orthogonal vibration $(E)\perp(B)\perp(k)$ $((k) \parallel (E) \times (B))$, phase coincidence, speed in vacuum $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = \frac{E_0}{B_0}$, speed in dielectric $v = \frac{c}{n}$.
Energetics & Radiation Pressure	Electromagnetic waves, their characteristics, and their transverse nature.	Equipartition of energy $(u_E = u_B)$, average energy density $u_{\text{avg}} = \frac{1}{2}\varepsilon_0 E_0^2 = \frac{B_0^2}{2\mu_0}$, intensity $I = u_{\text{avg}}c$, Poynting vector (S), radiation pressure $P_{\text{rad}} = \frac{I}{c}$ (absorption) and $2\frac{I}{c}$ (reflection).
Electromagnetic Spectrum Taxonomy	Electromagnetic spectrum (radio waves, microwaves, infrared, visible, ultraviolet, X-rays, and gamma rays).	Wavelength and frequency orders $(\lambda_\gamma < \lambda_X < \lambda_{\text{UV}} < \lambda_{\text{Vis}} < \lambda_{\text{IR}} < \lambda_{\text{Micro}} < \lambda_{\text{Radio}})$, photon energy $E = h\nu = h\frac{c}{\lambda}$.
Production, Detection & Practical Uses	Electromagnetic spectrum including elementary facts about their uses.	Generation mechanisms (Coolidge tube X-rays, magnetron microwaves, molecular IR, nuclear gamma), detection tools, greenhouse effect, RADAR, LASIK, water

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Sub-topic	Official syllabus pointer (NTA Physics-322, Unit 5)	Key Exam Focus & Skills
		purifiers, cancer radiotherapy.

EXAM PATTERN

CUET UG Exam Architecture: Unit 5 accounts for roughly 6% to 8% of the CUET UG Physics test (3 to 4 MCQs). Standard question archetypes include identifying spectral bands by wavelength/frequency, calculating displacement current during capacitor charging, finding E_0 or B_0 using $c = \frac{E_0}{B_0}$, and radiation pressure / intensity relations.

Concept of Displacement Current & Ampere-Maxwell Law

Inconsistency of Ampere's Circuital Law

Ampere's Circuital Law states that the line integral of magnetic field (B) around any closed loop is equal to μ_0 times the total conduction current enclosed:

$$\oint (B) \cdot d(l) = \mu_0 I_c$$

In 1864, James Clerk Maxwell pointed out that this law is mathematically inconsistent for time-varying electrical processes, such as charging or discharging a parallel-plate capacitor:

- Consider a loop C encircling the wire feeding a charging capacitor.
- For a flat planar surface S_1 spanning loop C, conduction current passes through it: $I_{\text{enclosed}} = I_c \Longrightarrow \oint (B) \cdot d(l) = \mu_0 I_c$.
- For a pot-shaped balloon surface S_2 spanning the same loop C but bulging between the capacitor plates, no conduction current crosses it ($I_c = 0$): $\oint (B) \cdot d(l) = 0$.
- Since both surfaces share the identical boundary curve C, Ampere's original circuital law yields a mathematical contradiction ($\mu_0 I_c = 0$), indicating an incomplete equation!

Maxwell's Resolution: The Displacement Current (I_d)

Maxwell reasoned that between the capacitor plates, although no real charges cross the vacuum gap, the electric field (E) is continuously changing with time as charge accumulates on the plates ($q = CV$).

- Let A be the plate area and $\sigma = \frac{q}{A}$ be the surface charge density.
- The electric field between the plates is:

$$E = \frac{\sigma}{\varepsilon_0} = \frac{q}{\varepsilon_0 A} \Longrightarrow \Phi_E = EA = \frac{q}{\varepsilon_0}$$

- Differentiating with respect to time:

$$\frac{d\Phi_E}{dt} = \frac{1}{\varepsilon_0}\frac{dq}{dt} = \frac{I_c}{\varepsilon_0} \Longrightarrow I_c = \varepsilon_0 \frac{d\Phi_E}{dt}$$

DISPLACEMENT CURRENT & AMPERE-MAXWELL LAW

Displacement Current:

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt} = \varepsilon_0 A \frac{dE}{dt} \quad \text{[SI Unit: Ampere (A)]}$$

Generalised Ampere-Maxwell Law:

$$\oint (B) \cdot d(l) = \mu_0 (I_c + I_d) = \mu_0 I_c + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$

Continuity of Electric Current:

- Outside the capacitor plates: Conduction current exists ($I = I_c$), while $I_d = 0$.
- Between the capacitor plates: Displacement current exists ($I = I_d$), while $I_c = 0$.
- Total current $I = I_c + I_d$ is continuous across every cross-section of the entire circuit.

WORKED EXAMPLE 1: DISPLACEMENT CURRENT IN A CHARGING CAPACITOR

Problem: A parallel-plate capacitor with circular plates of radius $R = 12$ cm is being charged by an external current. At a certain instant, the electric field between the plates is changing at the rate of $d\frac{E}{d}t = 5.0 \times 10^{12}$ V/(m · s). Find: (a) the displacement current between the plates, and (b) the induced magnetic field at a distance $r = 6.0$ cm from the central axis.

Solution: Plate area: $A = \pi R^2 = \pi (0.12)^2 = 1.44\pi \times 10^{-2}$ m^2 $\approx 4.524 \times 10^{-2}$ m^2. (a) Displacement current:

$$I_d = \varepsilon_0 A \frac{dE}{dt} = (8.854 \times 10^{-12}) \times (4.524 \times 10^{-2}) \times (5.0 \times 10^{12}) = 2.00 \text{ A}$$

(b) Magnetic field at radius $r < R$: By the Ampere-Maxwell law, the displacement current enclosed within radius r is:

$$I_{d'} = I_d \left(\frac{\pi r^2}{\pi R^2} \right) = I_d \left(\frac{r^2}{R^2} \right) = 2.0 \times \left(\frac{(0.06)^2}{(0.12)^2} \right) = 2.0 \times \left(\frac{1}{4} \right) = 0.50 \text{ A}$$

$$B(2\pi r) = \mu_0 I_{d'} \Longrightarrow B = \frac{\mu_0 I_d r}{2\pi R^2} = \frac{(4\pi \times 10^{-7}) \times 0.50}{2\pi \times 0.06} = \frac{10^{-7}}{0.06} = 1.67 \times 10^{-6} \text{ T} = 1.67 \mu\text{T}$$

Maxwell's Equations & Hertz's Verification

The Four Fundamental Maxwell's Equations

Maxwell unified all classical electrostatic, magnetostatic, and electrodynamic phenomena into four elegant equations:

Law	Mathematical Statement	Physical Significance
1. Gauss's Law (Electrostatics)	$\oint (E) \cdot d(A) = \frac{q_{\text{enc}}}{\varepsilon_0}$	Electric charges are monopolar sources or sinks of electric fields.
2. Gauss's Law (Magnetism)	$\oint (B) \cdot d(A) = 0$	Magnetic field lines form continuous closed loops; isolated magnetic monopoles do not exist.

Law	Mathematical Statement	Physical Significance
3. Faraday-Lenz Law	$\oint (E) \cdot d(l) = -\frac{d\Phi_B}{dt}$	A time-varying magnetic flux induces a non-conservative, circulating electric field.
4. Ampere-Maxwell Law	$\oint (B) \cdot d(l) = \mu_0 \left(I_c + \varepsilon_0 \frac{d\Phi_E}{dt} \right)$	Magnetic fields are generated by electric conduction currents and time-varying electric fluxes.

Production of Electromagnetic Waves

- **Static Charge:** Produces only a static electrostatic field (E). No magnetic field $((B) = 0)$. No radiation.
- **Uniformly Moving Charge (Steady Current):** Produces both a steady electric field and a steady magnetic field. Energy is transported along conductors, but no electromagnetic radiation is emitted into space.
- **Accelerated Charge (Oscillating Dipole):** Produces a time-varying electric field $(E)(t)$, which generates a time-varying magnetic field $(B)(t)$, which in turn regenerates $(E)(t)$. This self-sustaining, mutually regenerating coupled oscillation decouples from the source and propagates through space as an **electromagnetic wave**.
- **Frequency of Radiation:** An oscillating charge radiating at frequency ν emits electromagnetic waves of the **identical frequency** ν.

Hertz's Experimental Verification

In 1887, Heinrich Hertz experimentally proved the existence of electromagnetic waves:

- **Transmitter:** An induction coil connected across two large polished brass spheres separated by a small spark gap. The oscillating spark acted as a high-frequency dipole oscillator ($\nu \approx 5 \times 10^7$ Hz, $\lambda \approx 6$ m).
- **Receiver:** An open metallic wire ring with tiny brass knobs creating a detector spark gap. Whenever sparks jumped across the transmitter gap, sympathetic sparks were observed across the receiver gap across the room, even without physical electrical connection!
- Hertz demonstrated that these waves exhibited reflection, refraction, interference, diffraction, and transverse polarization, travelling at the speed of light c.
- Later, Jagadish Chandra Bose generated extremely short millimetre waves (5 mm – 25 mm), and Guglielmo Marconi achieved transatlantic wireless radio transmission.

WORKED EXAMPLE 2: MAXWELL'S SOURCE AND FREQUENCY OF RADIATION

Problem: An electron in an atom oscillates about its equilibrium position with a frequency of 10^{14} Hz. What is: (a) the frequency, and (b) the wavelength of the electromagnetic waves emitted by the oscillating electron in vacuum?

Solution: (a) Frequency of emitted electromagnetic wave: According to Maxwell's theory, the frequency of electromagnetic waves emitted by an accelerated oscillating charge is identical to the mechanical frequency of oscillation:

$$\nu = 10^{14} \text{ Hz}$$

(b) Wavelength of the wave in vacuum:

$$\lambda = \frac{c}{\nu} = \frac{3.0 \times 10^8}{10^{14}} = 3.0 \times 10^{-6} \text{ m} = 3.0\mu\text{m}$$

This emitted wavelength falls in the **infrared** band of the electromagnetic spectrum.

Transverse Nature & Mathematical Description of EM Waves

Plane Electromagnetic Waves in Vacuum

Consider an electromagnetic wave propagating along the $+x$-direction. The electric field oscillates sinusoidally along the y-axis, and the magnetic field oscillates sinusoidally along the z-axis:

SINUSOIDAL WAVE EQUATIONS

$$E_{y(x,t)} = E_0 \sin(kx - \omega t)$$

$$B_{z(x,t)} = B_0 \sin(kx - \omega t)$$

where:

- E_0 and B_0 are the peak amplitudes of the electric and magnetic fields.
- $k = \frac{2\pi}{\lambda}$ is the wave propagation constant (propagation vector).
- $\omega = 2\pi\nu$ is the angular frequency.
- $c = \frac{\omega}{k} = \nu\lambda = 3.0 \times 10^8$ m/s.

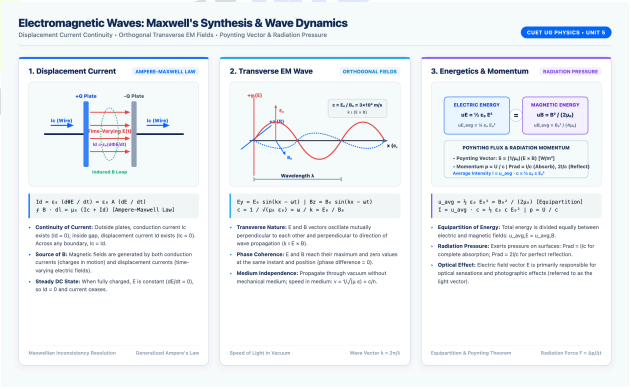


Figure 1: Electromagnetic Wave Propagation: Displacement current in capacitor gaps, transverse orthogonal field vectors ($E \perp B \perp k$), and energy equipartition.

Crucial Characteristics of Electromagnetic Waves

1. **Transverse Nature:** The electric field (E) and magnetic field (B) vibrate perpendicular to each other and perpendicular to the direction of propagation (k):
$$(k) \parallel ((E) \times (B))$$
2. **In-Phase Oscillations:** The electric and magnetic fields attain their maxima, minima, and nodal zeros at the exact same coordinates in space and moments in time. There is **zero phase difference** between (E) and (B).
3. **Ratio of Field Amplitudes:** At every point and instant, the ratio of electric field magnitude to magnetic field magnitude equals the speed of light:

$$\frac{E}{B} = \frac{E_0}{B_0} = \frac{E_{\text{rms}}}{B_{\text{rms}}} = c$$

4. **Velocity in Vacuum:** Determined purely by the universal electromagnetic constants:
$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 3.0 \times 10^8 \text{ m/s}$$
5. **Velocity in a Material Medium:** In a medium with permittivity $\varepsilon = \varepsilon_r \varepsilon_0$ and permeability $\mu = \mu_r \mu_0$:
$$v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{1}{\sqrt{(\mu_r\mu_0)(\varepsilon_r\varepsilon_0)}} = \frac{c}{\sqrt{\mu_r\varepsilon_r}} = \frac{c}{n}$$
where $n = \sqrt{\mu_r \varepsilon_r}$ is the refractive index of the medium (for non-magnetic media, $\mu_r \approx 1 \Longrightarrow n \approx \sqrt{\varepsilon_r}$).
6. **Light Vector:** The electric field vector (E) produces all visual, photographic, and physiological optical sensations because electric forces on charges ($F_e = qE$) are vastly stronger than magnetic forces ($F_m = qvB = qv\left(\frac{E}{c}\right) = \left(\frac{v}{c}\right)F_e \ll F_e$).

WORKED EXAMPLE 3: EM WAVE PARAMETERS AND FIELD EQUATION

Problem: A plane electromagnetic wave travels in vacuum along the x-axis. At a point, its electric field is $E_y = 6.0$ V/m directed along the $+y$-direction at frequency $\nu = 2.0 \times 10^7$ Hz. Determine: (a) the magnitude and direction of magnetic field (B), (b) the wavelength λ, and (c) the propagation constant k.

Solution: (a) Amplitude of magnetic field:

$$B_0 = \frac{E_0}{c} = \frac{6.0}{3.0 \times 10^8} = 2.0 \times 10^{-8} \text{ T} = 20 \text{ nT}$$

Since wave propagates along $+\hat{\imath}$ and (E) points along $+\hat{\jmath}$, by (k) $\parallel$ (E) $\times$ (B):

$$\hat{\imath} = \hat{\jmath} \times \hat{k} \Longrightarrow (B) \text{ must be oriented along the } +z\text{-direction } \left(+\hat{k}\right).$$

(b) Wavelength:

$$\lambda = \frac{c}{\nu} = \frac{3.0 \times 10^8}{2.0 \times 10^7} = 15 \text{ m}$$

(c) Propagation constant:

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{15} \approx 0.419 \text{ rad/m}$$

Energy Density, Poynting Vector & Radiation Pressure

Energy Density & Equipartition Principle

Electromagnetic waves transport energy through space. The total instantaneous energy density u is the sum of electric energy density u_E and magnetic energy density u_B:

$$u = u_E + u_B = \frac{1}{2}\varepsilon_0 E^2 + \frac{B^2}{2\mu_0}$$

AVERAGE ENERGY DENSITY & EQUIPARTITION

Taking time averages over a full cycle ($\sin^2(k\bar{x} - \omega t) = \frac{1}{2}$):

$$u_{E,\text{ avg}} = \frac{1}{4}\varepsilon_0 E_0^2, \quad u_{B,\text{ avg}} = \frac{B_0^2}{4\mu_0}$$

Substituting $B_0 = \frac{E_0}{c}$ and $c^2 = \frac{1}{\mu_0\varepsilon_0}$:

$$u_{B,\text{ avg}} = \frac{\left(\frac{E_0}{c}\right)^2}{4\mu_0} = \frac{E_0^2}{4\mu_0\left(\frac{1}{\mu_0\varepsilon_0}\right)} = \frac{1}{4}\varepsilon_0 E_0^2 = u_{E,\text{ avg}}$$

Equipartition Law: The total energy of an electromagnetic wave is divided **identically in half** between the electric and magnetic fields! **Total Average Energy Density:**

$$u_{\text{avg}} = u_{E,\text{ avg}} + u_{B,\text{ avg}} = \frac{1}{2}\varepsilon_0 E_0^2 = \frac{B_0^2}{2\mu_0} = \varepsilon_0 E_{\text{rms}}^2$$

Intensity of Electromagnetic Waves (I)

The intensity I is the average electromagnetic energy flowing per unit time per unit cross-sectional area perpendicular to the direction of propagation:

$$I = \frac{\text{Energy}}{\text{Area} \times \text{Time}} = \frac{u_{\text{avg}} \times A \times c \times \Delta t}{A \times \Delta t} = u_{\text{avg}} c$$

$$I = \left(\frac{1}{2}\varepsilon_0 E_0^2\right) c = \frac{1}{2}\varepsilon_0 c E_0^2 = \frac{E_0 B_0}{2\mu_0} = \frac{E_{\text{rms}} B_{\text{rms}}}{\mu_0}$$

Poynting Vector ((S))

The instantaneous rate of energy flow per unit area is represented by the **Poynting Vector**:

$$(S) = \frac{1}{\mu_0}((E) \times (B)) \quad [\text{SI Unit: W/m}^2]$$

- The direction of (S) specifies the instantaneous direction of wave energy propagation.
- The time average of Poynting vector magnitude equals the intensity: $|(\bar{S})| = I$.

Momentum & Radiation Pressure (P_{rad})

Electromagnetic waves carry linear momentum p. When radiant energy U strikes a surface:

- **Linear Momentum:** $p = \frac{U}{c}$.
- **Radiation Force:** $F = \frac{\Delta p}{\Delta t}$.
- **Radiation Pressure:** $P_{\text{rad}} = \frac{F}{A}$.

1. Completely Absorbing Surface: Wave is fully absorbed ($\Delta p = p = \frac{U}{c}$):

$$P_{\text{rad}} = \frac{F}{A} = \frac{\Delta \frac{p}{\Delta} t}{A} = \frac{\frac{\frac{U}{c}}{\Delta} t}{A} = \frac{I}{c}$$

$$P_{\text{rad}} = \frac{I}{c} = u_{\text{avg}} \quad [\text{N/m}^2 = \text{Pa}]$$

2. Completely Reflecting Surface: Wave is reflected back ($\Delta p = p - (-p) = 2p = 2\frac{U}{c}$):

$$P_{\text{rad}} = \frac{2I}{c} = 2u_{\text{avg}}$$

Radiation pressure on a perfect mirror is **twice** that on a black absorber!

WORKED EXAMPLE 4: SOLAR INTENSITY, FIELD AMPLITUDES & RADIATION FORCE

Problem: The solar constant (intensity of sunlight reaching Earth's upper atmosphere) is $I = 1380\ \text{W/m}^2$. Calculate: (a) the RMS electric field E_{rms}, (b) the RMS magnetic field B_{rms}, and (c) the total radiation force exerted on a perfectly reflecting solar sail of area $A = 100\ \text{m}^2$.

Solution: (a) RMS electric field:

$$I = \varepsilon_0 c E_{\text{rms}}^2 \Longrightarrow E_{\text{rms}} = \sqrt{\frac{I}{\varepsilon_0 c}} = \sqrt{\frac{1380}{(8.854\times10^{-12})\times(3.0\times10^{8})}} = \sqrt{\frac{1380}{2.656\times10^{-3}}} = \sqrt{5.195\times10^{5}} \approx 720.8\ \text{V/m}$$

(b) RMS magnetic field:

$$B_{\text{rms}} = \frac{E_{\text{rms}}}{c} = \frac{720.8}{3.0\times10^{8}} = 2.40\times10^{-6}\ \text{T} = 2.40\mu\text{T}$$

(c) Total radiation force on perfectly reflecting surface:

$$F = P_{\text{rad}} A = \left(\frac{2I}{c}\right) A = \frac{2\times1380\times100}{3.0\times10^{8}} = \frac{276000}{3.0\times10^{8}} = 9.20\times10^{-4}\ \text{N} = 0.920\ \text{mN}$$

The Complete Electromagnetic Spectrum

Orderly Taxonomy of Electromagnetic Radiation

The electromagnetic spectrum comprises the entire continuous range of electromagnetic frequencies, spanning over 24 orders of magnitude.

Name	Wavelength (λ)	Frequency (ν)	Production Method
Radio waves	> 0.1 m (up to 10^4 m)	$< 3\times10^9$ Hz (from 500 kHz)	Rapid acceleration of charges across oscillating LC electronic circuits and transmitting antennas.
Microwaves	1 mm – 0.1 m	$3\times10^9 - 3\times10^{11}$ Hz	Special vacuum tubes such as Klystrons, Magnetrons, and semiconductor Gunn diodes.
Infrared (IR)	700 nm – 1 mm	$3\times10^{11} - 4\times10^{14}$ Hz	Vibrations and rotations of atoms and molecules in hot incandescent bodies.
Visible light	400 nm – 700 nm	$4\times10^{14} - 7.5\times10^{14}$ Hz	Radiative de-excitation of outer valence electrons in excited atoms.
Ultraviolet (UV)	1 nm – 400 nm	$7.5\times10^{14} - 3\times10^{17}$ Hz	Inner shell electron transitions in very hot bodies, carbon/mercury electric arcs, and Sun.
X-rays	10^{-3} nm – 1 nm (0.01 Å – 100 Å)	$3\times10^{17} - 3\times10^{20}$ Hz	Deceleration of energetic cathode rays bombard-

Name	Wavelength (λ)	Frequency (ν)	Production Method
			ing heavy metal targets (Coolidge tube).
Gamma (γ) rays	$< 10^{-3}$ nm (< 0.01 Å)	$> 3 \times 10^{20}$ Hz	Nuclear radioactive transitions and de-excitation of excited atomic nuclei.

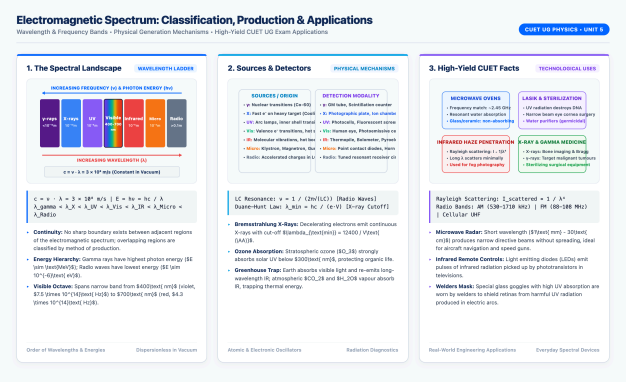


Figure 2: Electromagnetic Spectrum Architecture: Wavelength ladder, physical generation mechanisms, and essential CUET technological applications.

Detailed Spectral Characteristics & Real-World Uses

1. Radio Waves:

- **AM (Amplitude Modulated):** 530 kHz – 1710 kHz.
- **Short Wave (SW):** 1.7 MHz – 30 MHz (ionospheric sky wave reflection).
- **FM (Frequency Modulated):** 88 MHz – 108 MHz.
- **Television Broadcast:** 54 MHz – 890 MHz.
- **Cellular Phones:** UHF band (800 MHz – 950 MHz).

2. Microwaves:

- Used in **RADAR** (Radio Detection and Ranging) for aircraft speed and navigation.
- **Microwave Ovens:** Magnetron generates ≈ 2.45 GHz microwaves matching the rotational resonant frequency of water molecules, causing efficient dielectric heating of food while ceramic/glass dishes remain cool.

3. Infrared Radiation ("Heat Waves"):

- Emitted by all hot objects; absorbed readily by water, CO_2, and organic bonds.
- Used in **night-vision goggles**, remote control LED transmitters, physical therapy, and satellite greenhouse temperature mapping.
- **Haze Penetration:** Long wavelength minimizes Rayleigh scattering ($I \propto \frac{1}{\lambda^4}$), making IR ideal for foggy landscape photography.

4. Visible Light:

- Narrow octave (400 nm – 700 nm: Violet, Indigo, Blue, Green, Yellow, Orange, Red).
- Detected by retinal photoreceptors (rods and cones).

5. Ultraviolet Radiation:

6. X-Rays:

- Diagnostic radiography, bone fractures, dental checks, CT scans.

- **Ozone Shield:** Stratospheric ozone layer (O_3) strongly absorbs harmful solar UV rays ($\lambda < 300$ nm).
- **Germicidal Action:** Destroys bacteria DNA; used in water purifiers.
- **LASIK Eye Surgery:** Excimer lasers emit precise UV pulses to re-sculpt human corneal curvature.
- **Welder's Goggles:** Special glass protects eyes from intense UV sparks.
- **Crystallography:** X-ray diffraction reveals lattice constants ($\lambda \approx d \approx 1$ Å).
- **Duane-Hunt Law:** Minimum cut-off wavelength:

$$\lambda_{\min} = \frac{hc}{eV} \approx \frac{12400}{V} \text{ Å}$$

7. Gamma (γ) Rays:

- Extreme penetrating capability; used in radiotherapy to destroy cancerous cells, food irradiation, and nuclear physics.

WORKED EXAMPLE 5: X-RAY TUBE CUTOFF & PHOTON ENERGY COMPARISON

Problem: An X-ray tube operates at an accelerating potential of $V = 50$ kV. Determine: (a) the minimum wavelength $\lambda_{\min}$ of the emitted continuous X-rays, and (b) the ratio of photon energy of these X-rays to that of visible light of wavelength $\lambda = 500$ nm.

Solution: (a) Cutoff wavelength by Duane-Hunt relation:

$$\lambda_{\min} = \frac{hc}{eV} = \frac{(6.626 \times 10^{-34}) \times (3.0 \times 10^{8})}{(1.602 \times 10^{-19}) \times (50 \times 10^{3})} = \frac{1.988 \times 10^{-25}}{8.01 \times 10^{-15}} = 2.48 \times 10^{-11} \text{ m} = 0.248 \text{ Å}$$

(b) Energy of X-ray photon:

$$E_X = eV = 50 \text{ keV} = 50000 \text{ eV}$$

Energy of visible photon ($\lambda = 500$ nm $= 5000$ Å):

$$E_{\text{vis}} = \frac{hc}{\lambda} = \frac{12400}{5000} \approx 2.48 \text{ eV}$$

Ratio of photon energies:

$$\frac{E_X}{E_{\text{vis}}} = \frac{50000}{2.48} \approx 20161 \approx 2.0 \times 10^{4}$$

An X-ray photon carries roughly 20,000 times more energy than a visible light photon!

Atmospheric Propagation of Electromagnetic Waves

Electromagnetic waves propagate through the atmosphere via three distinct mechanisms depending on their frequency band:

1. Ground (Surface) Wave Propagation

- Ground waves travel over the Earth's surface, guided by the curved conducting boundary of the ground.
- **Attenuation:** The electric field induces alternating currents in the ground, causing Joule heat loss (I^2R) that increases dramatically with frequency (Loss $\propto f^2$).
- **Frequency Limit:** Limited to frequencies below 2 MHz ($f < 1500$ kHz $-$ 2000 kHz), primarily utilized for AM medium-wave radio broadcast.

2. Sky Wave (Ionospheric) Propagation

- Electromagnetic waves in the frequency range of 2 MHz to 30 MHz (short-wave band) are reflected back to Earth by total internal reflection in the ionized layers of the upper atmosphere (D, E, F1, F2 layers).
- **Critical Frequency (f_c):** The highest frequency of radio waves that can be reflected straight back from an ionospheric layer at vertical incidence:

$$f_c = 9\sqrt{N_{\max}} \quad [\text{Hz}]$$

where $N_{\max}$ is the maximum electron density in the layer (in electrons/m^3). Waves with $f > f_c$ penetrate through the ionosphere into space!
- **Maximum Usable Frequency (MUF):** For waves incident at an angle of incidence θ:

$$f_{\text{MUF}} = \frac{f_c}{\cos}\theta = f_c \sec\theta$$

- **Skip Distance (d_{skip}):** The minimum distance from the transmitter along the Earth's surface at which sky waves of a given frequency return to Earth.

3. Space Wave (Tropospheric / Line-of-Sight) Propagation

- Waves with frequencies greater than 30 MHz (30 MHz – 300 GHz: television, FM radio, radar, satellite, and cellular communication) pass straight through the ionosphere without reflection.
- Propagation occurs as a direct wave travelling in a straight line from transmitting antenna to receiving antenna within the troposphere.

LINE-OF-SIGHT (LOS) TRANSMISSION RANGE

For a transmitting antenna of height h_T and a receiving antenna of height h_R above the Earth's surface (Earth's radius $R \approx 6400$ km):

$$d_{\max} = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

- **Single Antenna Range:** $d = \sqrt{2Rh_T}$.
- **Surface Area Covered:** $A = \pi d^2 = \pi(2Rh_T) = 2\pi Rh_T$.
- **Population Covered:** Population = Population Density $\times$ $(2\pi Rh_T)$.

WORKED EXAMPLE 6: TV TRANSMISSION TOWER RANGE AND COVERAGE

Problem: A television transmitting antenna has a height of $h_T = 160$ m. If the receiving antenna is on the ground ($h_R = 0$) and the radius of the Earth is $R = 6.4 \times 10^6$ m, find: (a) the maximum distance up to which TV transmission can be directly received, and (b) the population covered if the average population density around the tower is 1200 persons/km^2.

Solution: (a) Maximum line-of-sight distance:

$$d = \sqrt{2Rh_T} = \sqrt{2 \times (6.4 \times 10^6) \times 160} = \sqrt{2.048 \times 10^9} = \sqrt{2048 \times 10^6} \approx 45.25 \times 10^3 \text{ m} = 45.25 \text{ km}$$

(b) Total surface area covered:

$$A = \pi d^2 = 2\pi Rh_T = 2\pi \times 6400 \times 0.160 = 2048\pi \approx 6434 \text{ km}^2$$

Total population reached:

$$\text{Population} = \text{Area} \times \text{Density} = 6434 \times 1200 \approx 7.72 \times 10^6 \text{ persons} = 77.2 \text{ lakhs}$$

Master Mindmap & High-Yield Cheatsheet



Figure 3: Master Synthesis Mindmap: Structural framework of Unit 5 Electromagnetic Waves.

MASTER ELECTROMAGNETIC WAVES REVISION CHEATSHEET

- **Displacement Current:** $I_d = \varepsilon_0\left(d\frac{\Phi_E}{d}t\right) = \varepsilon_0 A\left(d\frac{E}{d}t\right)$ | Ampere-Maxwell: $\oint(B)\cdot d(l) = \mu_0(I_c + I_d)$.
- **Current Continuity:** In capacitor circuit, $I_c = I_d$ at all times; inside capacitor gap, magnetic field $B(r) = \frac{\mu_0 I_d r}{2\pi R^2}$ for $r \le R$.
- **Transverse Waves:** $(E)\perp(B)\perp(k)$ $((k) \parallel (E)\times(B))$ | Speed $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = \frac{E_0}{B_0} = \frac{\omega}{k}$.
- **Refractive Index:** $n = \frac{c}{v} = \sqrt{\mu_r\varepsilon_r} \approx \sqrt{\varepsilon_r}$ (for non-magnetic media).
- **Equipartition of Energy:** $u_E = u_B = \frac{1}{4}\varepsilon_0 E_0^2$ | Total average $u_{\text{avg}} = \frac{1}{2}\varepsilon_0 E_0^2 = \frac{B_0^2}{2\mu_0}$.
- **Intensity & Poynting Vector:** $I = u_{\text{avg}}c = \frac{1}{2}\varepsilon_0 c E_0^2$ | $(S) = \left(\frac{1}{\mu_0}\right)((E)\times(B))$.
- **Radiation Pressure:** Absorbing: $P_{\text{rad}} = \frac{I}{c}$ | Perfectly reflecting: $P_{\text{rad}} = \frac{2I}{c}$.
- **Wavelength Order:** $\lambda_\gamma(< 10^{-12}\text{ m}) < \lambda_X(10^{-10}\text{ m}) < \lambda_{\text{UV}}(10^{-8}\text{ m}) < \lambda_{\text{Vis}}(400-700\text{ nm}) < \lambda_{\text{IR}}(10^{-5}\text{ m}) < \lambda_{\text{Micro}}(10^{-2}\text{ m}) < \lambda_{\text{Radio}}(> 0.1\text{ m})$.
- **Frequency Order:** $\nu_\gamma > \nu_X > \nu_{\text{UV}} > \nu_{\text{Vis}} > \nu_{\text{IR}} > \nu_{\text{Micro}} > \nu_{\text{Radio}}$.
- **Ozone Shielding:** Absorbs UV radiation below 300 nm; Greenhouse effect caused by selective IR trapping by atmospheric CO_2 and H_2O.

CUET UG Exam-Style Practice Questions (15 MCQs)

Q1. A parallel-plate capacitor with plate area A and plate separation d is charged by a current I. The displacement current density between the plates is:

(A) $\frac{I}{A}$ (B) $\varepsilon_0\left(\frac{I}{A}\right)$
(C) $\frac{Id}{A}$ (D) 0

Sol. Q1. By current continuity, total displacement current inside the dielectric gap equals the conduction current: $I_d = I_c = I$. The displacement current density is $J_d = \frac{I_d}{A} = \frac{I}{A}$.

Answer: (A)

Q2. Which of Maxwell's equations implies the non-existence of isolated magnetic monopoles in nature?

(A) $\oint(E)\cdot d(A) = \frac{q}{\varepsilon_0}$ (B) $\oint(B)\cdot d(A) = 0$
(C) $\oint(E)\cdot d(l) = -\frac{d\Phi_B}{d}t$ (D) $\oint(B)\cdot d(l) = \mu_0(I_c + I_d)$

Sol. Q2. $\oint(B)\cdot d(A) = 0$ (Gauss's Law for Magnetism) signifies that net magnetic flux over any closed Gaussian surface is zero, confirming magnetic monopoles do not exist.

Answer: (B)

Q3. The electric field of a plane electromagnetic wave in vacuum is $E_y = 30\sin(1.8x + 5.4\times 10^8 t)$ V/m. The wavelength of the wave is:

(A) 1.8 m (B) 3.49 m
(C) 2.50 m (D) 0.56 m

Sol. Q3. Comparing with $E_y = E_0\sin(kx+\omega t)$, the wave propagation constant is $k = 1.8$ rad/m. $\lambda = \frac{2\pi}{k} = \frac{2\pi}{1.8} = 3.49$ m.

Answer: (B)

Q4. In an electromagnetic wave in free space, the root-mean-square value of the electric field is $E_{\text{rms}} = 720$ V/m. The peak value of the magnetic field B_0 is:

(A) 2.4×10^{-6} T (B) 3.4×10^{-6} T
(C) 1.7×10^{-6} T (D) 4.8×10^{-6} T

Sol. Q4. $E_0 = \sqrt{2}E_{\text{rms}} = 1.414\times 720 = 1018$ V/m. Peak magnetic field $B_0 = \frac{E_0}{c} = \frac{1018}{3.0\times10^8} = 3.39\times10^{-6}$ T $\approx 3.4\times10^{-6}$ T.

Answer: (B)

Q5. The average electric energy density u_E and average magnetic energy density u_B in an electromagnetic wave travelling in vacuum satisfy:

(A) $u_E = 2u_B$ (B) $u_E = u_B$
(C) $u_B = 2u_E$ (D) $u_E = cu_B$

Sol. Q5. By the equipartition principle of electromagnetic waves, energy is equally divided between electric and magnetic fields, so $u_E = u_B = \frac{1}{4}\varepsilon_0 E_0^2$.

Answer: (B)

Q6. An electromagnetic wave of intensity $I = 600\ \text{W/m}^2$ falls normally on a completely reflecting flat mirror of area $0.50\ \text{m}^2$. The radiation force exerted on the mirror is:

(A) 1.0×10^{-6} N **(B)** 2.0×10^{-6} N
(C) 4.0×10^{-6} N **(D)** 8.0×10^{-6} N

Sol. Q6. Radiation pressure on a perfectly reflecting mirror is $P_{\text{rad}} = \frac{2I}{c}$. Force $F = P_{\text{rad}} A = \left(\frac{2I}{c}\right) A = \frac{2 \times 600 \times 0.50}{3.0 \times 10^8} = \frac{600}{3.0 \times 10^8} = 2.0 \times 10^{-6}$ N.

Answer: (B)

Q7. Which of the following electromagnetic radiations has the shortest wavelength?

(A) Microwaves **(B)** Ultraviolet rays
(C) X-rays **(D)** Gamma rays

Sol. Q7. Gamma rays have the highest frequency ($\nu > 10^{20}$ Hz) and consequently the shortest wavelength ($\lambda < 10^{-12}$ m).

Answer: (D)

Q8. The radiation used in RADAR systems for detection and navigation of aircraft is:

(A) Infrared rays **(B)** Microwaves
(C) Ultraviolet rays **(D)** Radio waves (short wavelength)

Sol. Q8. Microwaves ($1\ \text{mm} - 0.1\ \text{m}$) are used in RADAR because their short wavelength allows transmission as concentrated narrow beams that do not diffract or spread significantly.

Answer: (B)

Q9. The greenhouse effect, which keeps the Earth's surface warm, is primarily due to the trapping of:

(A) Solar ultraviolet rays by stratospheric ozone **(B)** Terrestrial infrared radiation by atmospheric gases like CO_2 and water vapour
(C) Solar visible light by atmospheric nitrogen **(D)** Cosmic gamma rays by the ionosphere

Sol. Q9. Earth re-radiates absorbed solar energy as long-wavelength infrared ("heat waves"), which are absorbed by CO_2 and water vapour in the lower atmosphere, trapping heat.

Answer: (B)

Q10. The minimum cut-off wavelength λ_{min} of continuous X-rays emitted by a Coolidge tube operating at accelerating voltage V is proportional to:

(A) V **(B)** $\sqrt{V}$
(C) $\frac{1}{V}$ **(D)** $\frac{1}{\sqrt{V}}$

Sol. Q10. By Duane-Hunt law, maximum kinetic energy of impacting electrons is converted into a single photon: $eV = h\nu_{\text{max}} = \frac{hc}{\lambda_{\text{min}}} \Longrightarrow \lambda_{\text{min}} = \frac{hc}{eV} \propto \frac{1}{V}$.

Answer: (C)

Q11. If ε_0 and μ_0 are electric permittivity and magnetic permeability of vacuum, and ε and μ are those in a medium, refractive index n of the medium is:

(A) $\sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}}$ **(B)** $\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}}$
(C) $\frac{\varepsilon\mu}{\varepsilon_0\mu_0}$ **(D)** $\frac{\varepsilon_0\mu_0}{\varepsilon\mu}$

Sol. Q11. $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$ and $v = \frac{1}{\sqrt{\mu\varepsilon}}$. Refractive index $n = \frac{c}{v} = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}} = \sqrt{\mu_r\varepsilon_r}$.

Answer: (A)

Q12. Which electromagnetic wave is produced when high-energy electrons are rapidly decelerated by bombarding a heavy metal target?

(A) Gamma rays **(B)** X-rays
(C) Ultraviolet rays **(D)** Microwaves

Sol. Q12. Energetic electrons decelerated in the Coulomb field of heavy target nuclei (e.g., Tungsten) emit Bremsstrahlung and characteristic X-rays.

Answer: (B)

Q13. The frequency band assigned to standard commercial FM radio broadcasting is:

(A) 530 kHz – 1710 kHz **(B)** 88 MHz – 108 MHz
(C) 54 MHz – 88 MHz **(D)** 800 MHz – 950 MHz

Sol. Q13. Commercial FM radio operates between 88 MHz and 108 MHz. AM broadcast operates between 530 kHz and 1710 kHz.

Answer: (B)

Q14. The Poynting vector (S) representing energy flux in an electromagnetic wave is given by:

(A) $(E) \times (B)$ **(B)** $\mu_0((E) \times (B))$
(C) $\left(\frac{1}{\mu_0}\right)((E) \times (B))$ **(D)** $\left(\frac{1}{\varepsilon_0}\right)((E) \times (B))$

Sol. Q14. The Poynting vector is defined as $(S) = \left(\frac{1}{\mu_0}\right)((E) \times (B))$, with SI units of Watts per square metre ($\mathrm{W/m^2}$).

Answer: (C)

Q15. Ultraviolet light is utilized in water purification systems primarily because:

(A) It ionizes minerals into essential salts **(B)** It induces fluorescence to illuminate impurities
(C) It penetrates deeply to heat water to boiling **(D)** It disrupts microbial DNA, destroying pathogens

Sol. Q15. UV-C radiation ($\lambda \approx 254$ nm) is absorbed by cellular DNA of bacteria and viruses, breaking molecular bonds and rendering pathogens inactive.

Answer: (D)